

$$\mu' = \frac{1}{2} \left[\frac{W_{01}}{Q_1} + \frac{A Ge \alpha_D \alpha_S}{m} \right]$$

$$\frac{N}{V} = \frac{C}{m}$$

$$\alpha_S = C_{0,DA} \epsilon_0 V_{DC} = 91,81 \times 8,854 \times 10^{-6} \times \frac{V_{DC}}{5V} \approx 4 \times 10^{-3} \frac{\mu N}{V} = 4 \times 10^{-9} \frac{N}{V}$$

$\frac{\mu N \cdot mm}{V^2 \cdot pf} \cdot \frac{pf}{mm} \cdot X = \frac{\mu N}{V}$

$$\alpha_D = 2\alpha_S = 8 \times 10^{-9} \frac{N}{V}$$

$$\mu' = -\frac{1}{2} \left[\frac{W_{01}}{Q_1} - \frac{A Ge \alpha_D \alpha_S}{m} \right] = -\frac{1}{2} \left[\frac{1}{m} \left(\frac{W_{01}}{Q_1} - A Ge \alpha_D \alpha_S \right) \right] = -\frac{1}{2} \frac{C + C_r}{m}$$

$$C_r = C \Rightarrow \mu' = -\frac{1}{2} \frac{2C}{m} = -\frac{C}{m} = 2\mu$$

$$\gamma' = \frac{\gamma}{2}$$



x dimezzare il Ringdown

$$A Ge \alpha_D \alpha_S = \frac{W_{01}}{Q_1} \text{ (mm)} \rightarrow m? \quad \rho = 2330 \text{ kg/m}^3$$

$$2 \text{ Beam} \begin{cases} L = 532 \mu m \\ \text{Sezione } 5 \times 24 \mu m^2 \end{cases} \quad m_b = 2,98 \times 10^{-10} \text{ kg}$$

$$\text{Struttura Centrale} \quad 27 \times 10 \times 24 \mu m^3 \quad m_c = 1,5 \times 10^{-11} \text{ kg}$$



$$m_T = m = 3,1 \times 10^{-10} \text{ kg}$$

MA Problema:

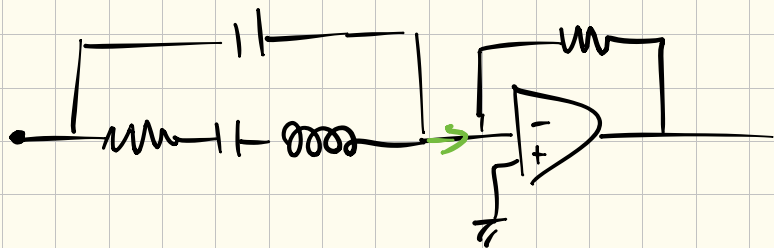
$q = 0$ spostamento del PAPER ma non c'è lo spostamento "reale"

spostamento reale $\mu = \psi q$ dove $\int \rho \psi^2 = M = 1$

non possiamo usare m_{reale} nell'equazione

in q

Fig 5 $\Rightarrow$ con $V_{DC} = 4V$ $i_0 \approx 6\mu A$
 $V_{AC} = 177,8mV$



$$m\ddot{x} + c\dot{x} + kx = \alpha_D V_{in}(t)$$

$$\frac{I_o(s)}{V_{in}(s)} = \frac{\alpha_D \alpha_S S}{mS^2 + cS + k}$$

Risonanza $\Rightarrow$ Massa e molla sovriscono

$$mS^2 + k = 0$$

$$\left. \frac{I_o(s)}{V_{in}(s)} \right|_{dis} = \frac{\alpha_D \alpha_S \cancel{S}}{c\cancel{S}} = \frac{\alpha_D \alpha_S}{c} = \frac{6 \times 10^{-9}}{0,178} \approx 3 \times 10^{-8} S = Y_{ris}$$

$$M' = -\frac{1}{2} \left[\frac{W_{01}}{Q_1} + \frac{A Ge \alpha_D \alpha_S}{m} \right]$$

$$\Rightarrow \text{da primo} \quad -\frac{1}{2} \left[\frac{C + C_r}{m} \right] \text{ dove } C_r = A Ge \alpha_D \alpha_S$$

$$C = m \frac{W_{01}}{Q_1}$$

$$M' = -\frac{1}{2} \frac{W_{01}}{Q_1} \left[1 + \frac{A Ge \alpha_D \alpha_S}{\frac{W_{01} m}{Q_1}} \right] = \frac{1}{2} \frac{W_{01}}{Q_1} \left[1 + \frac{A Ge \alpha_D \alpha_S}{C} \right]$$

$$\stackrel{!}{=} \frac{1}{2} \frac{W_{01}}{Q_1} \left[1 + A Ge Y_{ris} \right]$$

$$A Ge Y_{ris} = 1$$

$$A = \frac{1}{Ge Y_{ris}} = \frac{1}{10^{-6} \cdot 10^{-8} \cdot 3} = \frac{1}{0,03} = 33$$